Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/05/2005
PI Name:	Benton, Eric Ph.D.		
Project Title:	BEAMS: Benchmark Evaluations and Analysis of Materials		
Division Name:	Physical Sciences		
Program/Discipline:	RADIATION SHIELDING		
Program/Discipline--Element/Subdiscipline:	RADIATION SHIELDING--Radiation shielding		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	eric@erilresearch.com	**Fax:**	FY 415-422-2469
PI Organization Type:	INDUSTRY	**Phone:**	415-386-0254
Organization Name:	ERIL Research, Inc.		
PI Address 1:	P.O.Box 150788		
PI Address 2:			
PI Web Page:			
City:	San Rafael	**State:**	CA
Zip Code:	94915-0788	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-UNSOL
Start Date:	12/01/2002	**End Date:**	09/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Lin, Zi-Wei	**Contact Phone:**	256 544 3056
Contact Email:	linzx@nsstc.uah.edu		
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (7/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gersey, Brad (Prairie View A&M University) Shavers, Mark (NASA JSC/Wyle Laboratories) Uchihori, Yukio (National Institute for Radiological Sciences, Japan) Yasuda, Nakahiro (National Institute for Radiological Sciences, Japan) Benton, Eugene (Physics Dept., University of San Francisco) Frank, Allen (Physics Dept., University of San Francisco)		
Grant/Contract No.:	NAG8-1899		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Space flight radiation detectors including the CR-39 Plastic Nuclear Track Detector (PNTD), Thermoluminescent Detector (TLD), the NASA JSC Tissue-Equivalent Proportional Counter (JSC-TEPC), and the Liulin Mobile Dosimetry Unit (MDU) are being used to meet the "Measurements and Verification of Radiation Transport Code Accuracy" requirement set forth in NASA Research Announcement NRA-01-OBPR-05 through the establishment a set of thick target benchmarks. It is imperative that spaceflight detectors be included in any comprehensive program of radiation transport code validation since it will ultimately be detectors of this type that will be used for astronaut dosimetry on missions to the moon and Mars. Each set of benchmark measurements includes LET/y spectra, dose and dose equivalent as functions of shielding thickness. Measurements are conducted at ground-based particle accelerators including the the NASA Space Radiation Laboratory (NSRL) at the Brookhaven National Laboratory (BNL) and the HIMAC heavy ion accelerator at the National Institute for Radiological Sciences (NIRS), Chiba, Japan, using proton and heavy-ion beams of energy commonly encountered in the galactic cosmic ray (GCR) and solar energetic particle (SEP) environments. Measurements are made behind a set of "standard" targets including Al, Cu, polyethylene and water that varying in thickness from 0.5 to >30 g/cm^2. Target fragmentation measurements are being carried out behind thick targets exposed to protons at the Loma Linda University Medical Center and to neutrons at the Los Alamos Neutron Science Center (LANSCE). To aid in the accurate simulation of the experimental measurements, each thick target exposure is thoroughly documented and modeled. In addition, the detector response of the PDS, JSC-TEPC and Liulin are being modeled. To determine the "true" value of inclusive fragmentation cross sections, particle yields and other dosimetric quantities, results will be compared with results from other ground-based measurements. These spaceflight instruments can also be used in in situ assessments of shielding materials aboard spacecraft or balloon-borne test beds and in accelerator-based assessments of novel shielding materials. 1. Establish a baseline set of thick target "benchmarks" for use in assessing the accuracy of space radiation transport codes used to characterize the radiation shielding properties of spacecraft materials. These benchmarks will take the form of ground-based measurements of energetic heavy ions, protons, and neutrons covering a significant portion of the galactic cosmic ray (GCR) and solar energetic particle (SEP) spectra. Measurements are performed behind targets of composition relevant to spacecraft shielding, including Al, Cu, polyethylene and water, in a thickness range between 0.5 and >30 g/cm^2. Ions/energies are chosen based on the priorities outlined in NASA Research Announcement NRA-01-OBPR-05, the needs of the transport code development community, and NASA Conference Publication CP-3390. The measurements are being made with instrumentation routinely used for astronaut dosimetry during spaceflight including CR-39 plastic nuclear track detectors (PNTDs), thermoluminescent detectors (TLDs), a Tissue Equivalent Proportional Counter of the same design currently in use by the NASA Johnson Space Center (JSC-TEPC), and the Liulin MDU portable Si dosimeter. 2. In order to establish the "true" value of particle fields and other dosimetric quantities, results from ground-based exposures of the PDS, the JSC-TEPC and Liulin are being compared with each other and with results from other ground-based and spaceflight radiation detectors including the heavy ion spectrometer used in thin target measurements of HZE projectile fragmentation cross sections and instruments participating in the ICCHIBAN space radiation detector intercomparison project. The response of each of the detectors is being modeled, including the differences between the LET spectrum as measured by CR-39 PNTDs and the Lineal Energy Transfer (y) spectrum measured by the JSC-TEPC. 3. A set of "standard" targets has been fabricated and otherwise made available for use in these measurements. To aid the transport code development community in using the benchmark thick target measurements, experiment geometry of each individual exposure and the response of each detector are modeled and each exposure is thoroughly documented. The selection of target materials is based on input from the transport code development community. Elemental and simple compound targets include aluminum, copper, polyethylene, and water. Individual targets of a given composition will range in thickness from 0.5 to >30 g/cm^2 so that measurements can be made as a function of shielding depth. 4. The ability of transport codes to adequately model proton- and neutron-induced target fragmentation and the transport of light ions (Z = 1, 2) and neutrons is being assessed through measurements of particle yield and angular distribution of proton- and neutron-induced target fragments from tissue-equivalent and other target materials. Experiments will be carried out using the 250 MeV proton synchrotron at Loma Linda University Medical Center (LLUMC) and the Los Alamos Neutron Science Center (LANSCE). Measurements will be made using a CR-39 PNTD and, in some cases, detector readout is conducted using an Atomic Force Microscope in order to resolve short-range (<0.25 um) recoils and to minimize contamination from target fragmentation produced in the detector, itself. This project will provide a set of benchmark measurements of the ability of relevant materials to shielding the GCR radiation environment and can be used by the space radiation transport code development community to validate their models.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Results may benefit validation of radiation transport and shielding models used in ground-based applications such as in radiation cancer therapy and high energy particle accelerator research.
Task Progress:	In summary of our progress on the BEAMS project, we have continued our campaign of making dosimetric measurements behind thick targets during heavy ion accelerator exposures using CR-39 PNTD, TEPC, and the Liulin-4 MDU for use in validating the radiation transport codes and have made considerable progress in the analysis of this data, especially that taken with CR-39 PNTD. Comparison of the response of space flight dosimeters has continued in conjunction with the ICCHIBAN project. We have continued to use thick targets of polyethylene, aluminum, copper and graphite, but have expanded this set to include water and lead. The target fragment studies have yielded surprising and very valuable results, namely the direct measurement of recoils from heavy target nuclei following high energy proton interactions. We are currently analyzing CR-39 PNTD, TLD and Al2O3:C optically stimulated luminescence (OSL) detectors exposed aboard the Russian Service Module of ISS in 2004 as part of a international intercomparison experiment.
Bibliography Type:	Description: (Last Updated: 11/13/2025)
Articles in Peer-reviewed Journals	DeWitt JM, Benton ER, Uchihori Y, Yasuda N, Benton EV, Frank AL. "Assessment of radiation shielding materials for protection of space crews using CR-39 plastic nuclear track detector." Radiat Meas. 2009 Oct-Nov;44(9-10):905-8. http://dx.doi.org/10.1016/j.radmeas.2009.10.041 , Oct-2009

Articles in Peer-reviewed Journals	Johnson CE, Benton ER, Yasuda N, Benton EV. "Analysis of short-range tracks and large track fluences in CR-39 PNTD using atomic force microscopy." Radiat Meas. 2009 Oct-Nov;44(9-10):742-5. http://dx.doi.org/10.1016/j.radmeas.2009.10.080 , Oct-2009
Articles in Peer-reviewed Journals	DeWitt JM, Benton ER. "Shielding effectiveness: A weighted figure of merit for space radiation shielding." Appl Radiat Isot. 2020 Jul;161:109141. https://doi.org/10.1016/j.apradiso.2020.109141 ; PMID: 32250841 , Jul-2020
Articles in Peer-reviewed Journals	DeWitt JM, Benton ER. "Secondary proton buildup in space radiation shielding." Life Sci Space Res. 2024 May;41:119-126. https://doi.org/10.1016/j.lssr.2024.02.005 ; PubMed PMID: 38670638 , May-2024
Presentation	Benton, E. R.; Yasuda, N., Frank, A. L.; Moyers, M. F.; and Benton, E. V. "Heavy Recoil Tracks from Proton- and Neutron-Induced Non-elastic Interactions Imaged in CR-39 PNTD" 16th Annual NASA Space Radiation Investigators' Workshop May-2005
Presentation	Akselrod, M. S.; Akselrod, G. M.; Nelson, G. A.; Benton, E. R.; Yasuda, N. "A Novel Fluorescent Nuclear Track Detector with Application to Radiobiology and Space Dosimetry" 16th Annual NASA Space Radiation Investigators' Workshop May-2005